IEEE Access
Multidisciplinary ⋮ Rapid Review ⋮ Open Access Journal

Received 23 March 2026, accepted 27 March 2026, date of publication 2 April 2026, date of current version 10 April 2026.

Digital Object Identifier 10.1109/ACCESS.2026.3680313

RESEARCH ARTICLE

Effects of Interaction and Explanation Type on Human–AI Collaboration for Fake News Detection

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This work was supported in part by the National Funds through Fundação para a Ciência e a Tecnologia, I.P. (FCT) under projects UID/50021/2025 and UID/PRR/50021/2025 and project CRAI under Grant CC628696807-00454142 (IAPMEI/PRR); in part by the PNRR-M4C2-Investimento 1.3, Partenariato Esteso-''FAIR-Future Artificial Intelligence Research''-Spoke 1 ''Human-Centered AI,'' funded by the European Commission through the NextGeneration EU Program under Grant PE00000013; in part by the TAILOR Connectivity Fund part of the TAILOR Project funded by the EU Horizon 2020 Research and Innovation Program under Grant 952215; and in part by the AMOR Project, funded by the Spanish Ministry of Economic Affairs and Digital Transformation and the European Union-NextGeneration EU, within the UNICO I+D Cloud Program under Grant TSI-063100-2022-0002.

ABSTRACT AI-assisted decision-making systems are increasingly used across domains, prompting research into design factors that influence their effectiveness. Prior work has examined elements such as explanation type, interface design, and cognitive support interventions; however, the complexity of real-world tasks makes the generalization of findings difficult. In particular, in real-world decision-making settings like *Fake News Detection* studies on human-AI collaboration remain underexplored. This study investigates how explanation type and system interactivity affect human-AI collaboration in the context of fake news detection. In a controlled online experiment ($N = 161$), we varied the explanation type and the interactivity of an AI system that assists with the fake news detection task. Results show that conversational systems increased user reliance, while feature importance explanations appeared to be more effective than counterfactual explanations. These findings highlight the importance of explanation design and interaction modality in shaping effective AI-assisted decision-making in real world settings where underlying regularities are difficult for non-expert users to detect.

INDEX TERMS AI-assisted decision making, human–AI interaction, XAI, fake news detection, conversational systems.

I. INTRODUCTION

With the advent of Artificial Intelligence (AI) technology, human decision-making is becoming a hybrid process, in which humans receive assistance from AI systems that provide recommendations to inform their final judgment [1]. The field of AI-assisted decision-making is expanding rapidly, and researchers are increasingly examining the factors that enable effective collaboration between human decision-makers and AI systems. [2]. Many factors influence the collaborative process and the final accuracy of decisions: human expertise in the decision task [3], [4], the presence or type of explanations provided by the AI system [3], [5], [6], [7], the design of the task [4], [8], and specific interventions to mitigate cognitive bias and improve human trust and confidence in AI systems [9], [10]. These factors have been investigated to understand the ideal balance, where humans neither over-rely on the system when they shouldn't nor under-rely on it when it is correct.

A significant challenge in this research area is that most of these factors and system designs are highly context-dependent [2]. The influence of a factor, such as the type of explanation or the AI system's design, can strongly depend on user expertise or task difficulty [3], [6]. Moreover,

The associate editor coordinating the review of this manuscript and approving it for publication was Jenny Mahoney.

most studies focus on high-stakes, high-expertise domains such as healthcare [4], [11], or on relatively intuitive, low-difficulty tasks such as credit assessment or housing price prediction [3], [5], [6].

Despite the growing adoption of AI systems in decision-making processes, there is a limited understanding of how people interact with AI during complex, realistic decision-making tasks, such as detecting fake news, identifying scams, or personal financial planning. These domains are frequently overlooked in the literature, likely because their high complexity places a heavy cognitive burden on lay users—the target demographic—which significantly increases the propensity to over-rely. Investigating human-AI interaction in these understudied contexts is therefore critical for two reasons. First, it informs the design of AI systems that are both trustworthy and accessible to everyday users. Second, given that these tasks often occur in real-world settings where AI assistance may not always be available, it is essential to understand whether interacting with an AI system produces a "carry-over effect"—that is, whether it helps users perform the task more effectively even after the AI support is removed.

To understand whether established patterns of human-AI interaction extrapolate to these complex environments, two pivotal factors require analysis: explainability (whether different explanation types function to enhance understanding and reduce over-reliance) [12], [13] and interactivity (whether the conversational modality improves collaboration outcomes) [14], [15].

With this in mind, we sought to better understand the dynamics of collaboration in AI-assisted decision-making scenarios. We manipulated the type of explanation provided to help users interpret the AI's output, and the level of interactivity in the system in a *Fake News Detection* task. Our goal was to investigate how these factors influence user interaction and decision performance, not only during the AI-assisted task, but also afterward, when users must make similar decisions independently. Specifically, we ask:

RQ1. How does the interactivity of the AI assistant affect the decision-making process and human–AI collaboration?

RQ2. How do different explanation types influence users' decision performance and their reliance patterns in the AI system?

RQ3. What are the carry over effects of AI system design on users' decision-making when they later perform the task without AI assistance?

To address these research questions, we conducted a user study (N = 161) in which participants completed a fake news detection task in two parts: first with the help of an AI assistant and then without any AI assistant support. We designed a 2×2 between-subjects experiment, manipulating two factors: the type of explanation provided by the AI system and its interactivity mode. For explanation type, we focused on two of the most commonly used approaches: (1) feature importance explanations, where the system presents the features that contributed most to its decision, and (2) counterfactual explanations, where the system presents a set of feature-based rules that would need to be true for it to change its prediction. For the interactivity mode, the AI system was either (a) a static system, where suggestions and explanations were presented in a fixed format, or (b) a conversational system, where the AI could respond to user input and provide additional clarifications or explanations interactively. Our investigation focused on users' performance in detecting fake news, their agreement patterns with AI recommendations, and their self-reported trust in the AI assistant.

Results reveal that user behavior, specifically detection accuracy and agreement with the AI assistant, varied depending on both the assistant's interactivity and the type of explanation. In the context of fake news detection, counterfactual explanations were less effective than feature-based ones. Moreover, while the conversational AI systems was more persuasive than the static one, it did not consistently improve decision accuracy. Additionally, the conversational system showed a carryover effect on the decision-making performance. Users who had access to the conversational system were more confident in their responses when completing the task independently. Finally, contrary to common assumptions that AI explanations and interface design primarily shape user trust, users' self-reported trust correlates more with individual traits, such as news consumption habits and prior AI familiarity, than with the system's explanations or interface design. Taken together, this work makes three key contributions:

1) It provides empirical evidence on how explanation types and interaction modalities influence human decision-making in fake news detection decision-making tasks;
2) It offers actionable design insights for developing AI tools that go beyond algorithmic accuracy to actively support user trust and engagement for real-world settings;
3) It highlights the role of user characteristics in shaping perceptions of AI systems.

II. RELATED WORK

A. FAKE NEWS DETECTION

The rapid proliferation of digital content has revolutionized how information is accessed and consumed. While this transformation has brought immense benefits with the dissemination and democratization of news, it has also led to the widespread dissemination of fake news, which poses a significant threat to the integrity of information ecosystems [16], [17]. Fake news refers to intentionally false or misleading information presented as legitimate news, often designed to manipulate emotions, influence public opinion, and undermine trust in credible sources [18]. Its impact extends across social, political, and economic domains, exacerbating polarization, distorting democratic processes, and eroding institutional trust.

In this work, we follow common distinctions in the misinformation literature. Misinformation refers to false or inaccurate information that is shared without an explicit intention to deceive, for example, due to misunderstanding or negligence [19]. Disinformation, in contrast, refers to false or misleading information that is created and disseminated deliberately to deceive, manipulate, or cause harm [19]. Within this spectrum, we use the term fake news to denote fabricated news articles that mimic the format and style of legitimate journalism while conveying factually false content [20].

1) AUTOMATED FAKE NEWS DETECTION SYSTEMS

Automated approaches to fake news detection have evolved considerably in recent years, driven by advancements in machine learning and natural language processing. These systems can be broadly categorized into content-based and social context-based methods. [21], [22].

Content-based methods analyze the intrinsic properties of news articles, such as linguistic patterns, stylistic features, and multimedia elements. Lexical features, including sentiment polarity and readability metrics, have shown effectiveness in identifying deceptive text [23]. At the same time, psycholinguistic markers like emotional intensity and syntactic anomalies are leveraged in style-based approaches to detect manipulation. Complementing these, visual methods focus on uncovering image manipulation artifacts [24]. In contrast, social context-based techniques examine how information spreads by modeling user engagement metrics, propagation dynamics, and network structures [25]. Prominent solutions have also combined both approaches to develop hybrid frameworks that simultaneously analyze content and contextual signals. For instance, graph neural networks that integrate textual features with propagation cascades have achieved state-of-the-art results in misinformation detection [26].

The evolution of machine learning has revolutionized fake news detection through automated feature learning. Early systems relied on traditional classifiers, such as logistic regression and SVMs, trained on handcrafted features such as lexical diversity and punctuation patterns [27]. The introduction of word embeddings, such as Word2Vec and GloVe, enabled semantic analysis at scale, improving the ability to recognize covert propaganda techniques [28]. Deep learning architectures further advanced detection capabilities by capturing hierarchical feature representations. CNNs excelled at local pattern detection in text and images, while RNNs modeled temporal dependencies in news propagation [29]. More recently, transformer-based models like BERT set new benchmarks in accuracy through self-attention mechanisms but faced challenges related to computational demands and interpretability [30].

Current solutions in fake news detection emphasize enhancing accuracy, robustness, and interpretability through diverse methodologies. For instance, researchers have developed logic-based models that incorporate interpretable reasoning processes for multimodal misinformation detection [31]. Additionally, graph-based approaches have been proposed to optimize network structures and provide explainable insights into classification decisions [26].

2) HUMAN-CENTERED APPROACHES TO COMBAT FAKE NEWS

In response to the growing threat of misinformation, a variety of systems have been developed to support users in identifying and evaluating fake news [13], [32]. These human-centered interventions are designed to help individuals engage more critically with digital content, often by augmenting their decision-making with accessible tools and cues.

One common approach is manual fact-check, typically offered through dedicated websites such as PolitiFact or Snopes [33]. These platforms allow users to search for specific claims and access evidence-based verdicts verified by experts. Another widely adopted strategy involves integrating warning labels and credibility notices directly into social media platforms. For example, users encountering disputed content may see notices like "disputed by fact-checkers" or links to third-party verification sources, which aim to reduce the spread of misinformation and encourage reflective judgment. Browser-based tools and extensions such as NewsGuard provide real-time credibility ratings for news websites. These tools function passively as users browse the internet, offering immediate visual indicators of a source's reliability. Their technical performance has improved significantly in recent years, with state-of-the-art models showing strong classification capabilities [34]. Even government and political institutions are increasingly investing in these technologies to confront the escalating threat of online misinformation [35]. Manual fact-checking by organizations such as PolitiFact has been shown to significantly reduce belief in false claims, with effects lasting up to two weeks [36]. Social media platforms have also introduced warning labels and credibility scores, which can reduce belief and sharing of false content [37]. Prior research has shown that even moderately accurate detection systems can positively influence users' judgments [38]. Furthermore, providing explanations for a system's output can improve both its perceived effectiveness and users' trust in the system [39].

This complex environment, where fake news detection relies on both automated systems and human judgment, offers a compelling scenario for studying AI-assisted decision-making. Ultimately, it is the end-user who must decide whether to believe a piece of news, even when supported by AI recommendations.

B. EXPLAINABLE AI

The increasing use of AI systems in high-stakes domains has intensified the demand for transparency, making Explainable AI (XAI) a crucial area of research [40]. While early approaches often contrasted interpretable-by-design models with post-hoc explanations, recent work proposes more nuanced taxonomies that classify XAI methods along multiple dimensions [41]. A common framework distinguishes methods by their scope, model dependency, and explanation form. In terms of scope, global methods seek to characterize the model's overall behavior, whereas local methods explain individual predictions. Model-agnostic techniques can be applied to any black-box model, while model-specific methods exploit architectural details, such as saliency maps in convolutional networks or attention in transformers [42]. Regarding form, explanations can be based on feature-based attributions, example-based reasoning, or surrogate approximations [41].

Among this diverse and expanding field of research, several additional families of XAI methods have gained prominence, each offering different interpretive perspectives on model behavior. Concept-based explanations seek to interpret models using higher-level, human-understandable concepts rather than raw features; for instance, TCAV quantifies a model's sensitivity to semantic concepts [43]. Counterfactual explanations, in contrast, focus on identifying minimal or meaningful changes to an input that would alter a model's prediction, with foundational work establishing their role in model transparency and actionability [44]. Another line of work involves prototype and criticism methods, which use representative and non-representative examples to reveal how models structure the data internally; MMD-critic is a prominent example of such approaches [45]. Disentangled and interpretable representation learning aims to enforce structure in latent spaces, with methods like β-VAE encouraging models to separate generative factors so that latent dimensions correspond to interpretable variations [46]. Finally, mechanistic interpretability investigates neural models at a fine-grained structural level, analyzing neurons, attention heads, or computational circuits to uncover how information flows and how specific subnetworks implement functional roles [47].

Beyond traditional technical modalities, narrative explanations and conversational systems have emerged as effective ways to bridge the gap between AI reasoning and human understanding. One promising approach is narrative-based explanations, which offer a more natural and comprehensive understanding of AI recommendations [48]. Another approach involves presenting the pros and cons of different options rather than explaining the AI's internal reasoning, helping users make more informed choices [49]. Nonetheless, these XAI systems, which are more human-centered and prioritize human-AI collaboration and effectiveness, are still in their beginning and far from being well understood in terms of their effects and efficiency.

Beyond the type and scope of explanation, evaluation, and usability have emerged as critical dimensions. There is growing recognition that explanations must be not only technically accurate but also meaningful and actionable for different user groups and different contexts [2], [50], [51]. The field thus increasingly emphasizes user-centered and task-specific evaluation frameworks, alongside technical metrics like fidelity and stability [41].

C. AI ASSISTANCE IN DECISION MAKING: APPLICATIONS TO FAKE NEWS AND CONVERSATIONAL SYSTEMS

AI-assisted decision-making is an increasingly prominent area within human–AI collaboration, where AI systems offer recommendations or predictions to support human users who remain responsible for the final decision [1]. This paradigm spans diverse domains, from high-stakes applications such as medical diagnosis [4], [11] to more casual consumer tasks such as bird or plant identification [7], [52], [53]. Despite its promise, three persistent challenges have been identified: (1) ensuring that the AI meaningfully complements human judgment, (2) enabling users to build accurate mental models of the AI's strengths and limitations, and (3) supporting effective interaction through transparent system design [1]. These challenges are well-documented but far from resolved, particularly as the complexity of AI systems—and their use contexts—continues to evolve [2].

A growing body of empirical work has begun to unpack these challenges by studying user interactions with AI across different decision-making scenarios [2]. Such studies examine decision outcomes and human perceptions and behaviours towards the AI, specifically trust, a subjective psychological state capturing the willingness to depend on the AI, and reliance, the behavioral act of following AI advice [3], [6], [7], [10], [54], [55]. While trust is often captured through self-reported measures because it is difficult to define objectively, reliance is frequently used as its behavioral proxy [12], [56]. Crucially, however, these two variables are not necessarily correlated, as users may rely on a system while reporting varying levels of trust [7], [57]. Many of these investigations into trust, reliance and human-AI collaboration, however, have taken place in contexts involving either high domain expertise (e.g., medical or financial professionals) [3], [4], [58] or tasks with intuitive answers and low stakes (e.g., image classification) [6], [52].

Less is known about user interaction with AI in complex, real-world domains, such as fake news detection, where the reasoning process is often non-intuitive, and users may lack specialized expertise.

In the fake news domain, human-centered research is still emerging. Some work has focused on identifying explanation needs for professional fact-checkers [32], while others have explored whether AI-provided warnings or feature-based explanations help users better evaluate misinformation [38], [39]. Findings suggest that certain explanation types (e.g., feature importance) can help improve decision accuracy [39], but there remains a limited understanding of how different explanation strategies compare.

Moreover, researchers in AI-assisted decision-making have recently turned to conversational XAI, interactive systems that use dialogue to provide explanations, often leveraging large language models (LLMs), to improve human-AI collaboration in the decision task. These systems aim to promote cognitive engagement and reduce blind reliance by encouraging users to explore, question, and deliberate [59], [60], [61]. For example, techniques have emerged that use in-context learning (ICL) with LLMs to generate tailored post-hoc explanations [62], or convert outputs from traditional explainability tools like SHAP and LIME into more human-readable formats via prompt engineering [63]. While promising, conversational XAI also introduces risks: findings suggest that natural, fluent language may increase user confidence without necessarily improving decision accuracy—raising concerns about over-reliance on persuasive systems [15].

Despite this momentum, two key gaps remain. First, many studies exploring explanations and conversational XAI have focused on either expert users or highly intuitive tasks [15], [64]. Less is known about how these design features affect user behavior in more ambiguous domains for non-expert, everyday users, like fake news detection. Second, while explanations and interaction design are often studied in isolation, there is limited research on their combined impact, not only during AI-assisted decision-making, but also after, when users must act independently.

This work addresses these gaps by systematically comparing explanation types and levels of interactivity in a fake news detection task. We examine how these design choices influence decision-making during collaboration and how their effects persist when users later make decisions without AI support. In doing so, we contribute to a deeper understanding of how explanation and interaction design jointly shape trust, performance, and human–AI collaboration in complex, real-world tasks.

III. USER STUDY

A. OVERVIEW

To address our research questions, we conducted a controlled user study to investigate how the interactivity of an AI system and the type of explanation it provides affect human-AI collaboration in a fake news detection decision-making task. The study involved two sequential tasks: one in which participants interacted with an AI system to classify news articles, and a second follow-up task in which they performed the classification without assistance.

B. EXPERIMENTAL DESIGN

We employed a 2×2 between-subjects design, manipulating two independent variables:

1) **Explanation Type:**
   - *Feature Importance Explanation:* The AI highlights which features contributed most to its classification.
   - *Counterfactual Explanation:* The AI explains what minimal changes to the article would have altered its decision.
2) **Interactivity Level:**
   - *Static System:* Explanations are shown in a fixed, non-interactive format.
   - *Conversational System:* Explanations are delivered via a chatbot-style interaction, allowing users to ask follow-up questions.

This resulted in four experimental conditions: (1) a **Conversational System with Counterfactual Explanation**, where the AI system appeared as a chatbot and provided counterfactual explanations, highlighting which changes in content would alter its classification; (2) a **Conversational System with Feature Importance Explanation**, where the AI system appeared as a chatbot and provided feature importance explanations, emphasizing the main factors influencing its decision; (3) a **Static System with Counterfactual Explanation**, where the AI system was presented in a static format, offering a non-interactive counterfactual explanation; and (4) a **Static System with Feature Importance Explanation**, where the AI system was presented in a static format, displaying a feature importance explanation with no interaction.

We focused on two of the most widely used XAI techniques: feature importance explanations and counterfactual explanations. While more advanced explanation methods exist, these two were chosen for their widespread adoption and versatility across domains, as discussed in Section II-B. Our manipulation of interactivity level reflects the evolution of current design paradigms [14], [15], [60]. The static explanation system represents the conventional format commonly found in existing AI systems and serves as a baseline. In contrast, the conversational system captures the emerging trend of interactive, language-model-based assistants, which are increasingly proposed to improve transparency, collaboration, and user engagement (see Section II-C). Notably, the study does not include a no-explanation control condition. This was a deliberate choice: the absolute benefit of providing AI explanations over offering none has been established in prior work [39], and the present study was specifically designed to compare explanation types and interaction modalities rather than to revisit whether explanations are beneficial per se.

C. TASK DESCRIPTION

1) FAKE NEWS DETECTION TASK

The *fake news detection* task involved analyzing short textual articles and determining whether each article was legitimate or fake. In our setting, fake articles are entirely fabricated news-style pieces written to resemble real journalism, but that describe events that did not actually occur. As such, the task operationalizes fake news detection as a binary disinformation-detection problem, in which participants must distinguish between legitimate news and fabricated disinformation articles. Each article consisted of a title and body (ranging from 300 to 600 words). The study included two variants of this task: the Categorization Task, in which an AI system was present, and the Evaluation Task, in which no AI support was provided. This task domain was chosen because it represents an everyday activity frequently encountered

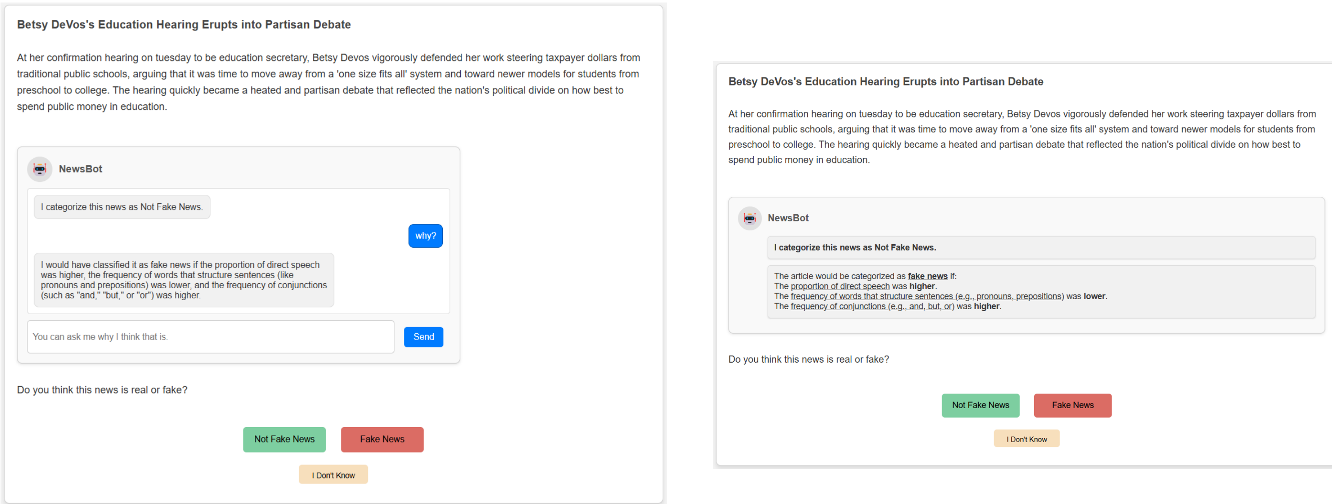

FIGURE 1. Fake news detection task – *Categorization Task*. The left image shows the conversational system condition featuring the chatbot version of NewsBot. The right panel shows a static system condition, with the NewsBot design, but with no opportunity for the user to interact with the AI system. Both panels depict the AI system providing counterfactual explanations.

by non-experts. While users may not always make explicit judgments about news veracity, they routinely engage in implicit credibility assessments as they consume information online. Importantly, this task is inherently challenging, involving incomplete context, making it a valuable testbed for evaluating AI-assisted decision-making in low-expertise, realistic settings.

2) CATEGORIZATION TASK (WITH AI SUPPORT)

In the **Categorization Task**, the article was displayed in full, and a categorization suggestion from an AI system, called NewsBot, was available. This suggestion was accompanied by an explanation of the AI's reasoning. Depending on the experimental condition, this explanation was either interactive (via a chatbot interface) or static (via plain text). The task interface allowed the user to choose among three classification options: *Fake News*, *Not Fake News*, or *I Don't Know*. Figure 1 illustrates the Categorization Task interface, both for the static and conversational conditions, and for the counterfactual type of explanations. In supplementary material there is an illustration of the categorization task interface for the feature importance explanations.

The decision to provide three response options, rather than the conventional binary choice of *Fake* or *Not Fake*, was made to better capture user reliance behaviors on the AI system. The most rigorous way to measure reliance is to record the user's initial judgment before seeing the AI recommendation, and then compare it to their revised judgment afterward, which reveals whether they change their mind and conform to the AI's advice [65]. However, this approach can disrupt the natural flow of a task and is less feasible in real-world applications. Instead, we opted for a setup that more closely mirrors real-world conditions, in which users see the AI recommendation upfront and make a single classification decision. Within this setting, the inclusion of the *I Don't Know* option offers a way to capture more nuanced user responses. It allows participants to express uncertainty without necessarily endorsing the AI's suggestion. When combined with a reward structure that penalizes incorrect answers and incentivizes correct ones (see Section IV-D), this third option becomes particularly meaningful: selecting *I Don't Know* indicates a lack of confidence and a choice to avoid committing, neither entirely relying on nor overriding the AI system.

3) EVALUATION TASK (WITHOUT AI SUPPORT)

In the **Evaluation Task**, a similar fake news classification task was presented, but without any AI support. The task flow remained the same: the article was shown in full, and a classification decision was required, with the same three possible responses. At the end of the task, users were asked to provide a free-text brief explanation of their classification decision. Figure 2 shows the interface used for the Evaluation Task. Importantly, manipulations were applied only during the Categorization Task. The Evaluation Task served as a follow-up measure to examine how prior exposure to different AI systems and explanations influenced users' subsequent independent decisions.

D. FAKE NEWS DETECTION SYSTEM

Following established methodologies in misinformation research [66], we frame fake news detection as a binary classification task. The system receives an article's text T and outputs a label $y \in \{0, 1\}$, where $y = 1$ indicates that the article is fake news.

We developed a model that combines interpretable feature engineering with a traditional machine learning classifier, prioritizing transparency without sacrificing predictive performance. Building on prior work that demonstrated the value of linguistic and psychological signals for deception detection [67], we analyzed five feature dimensions:

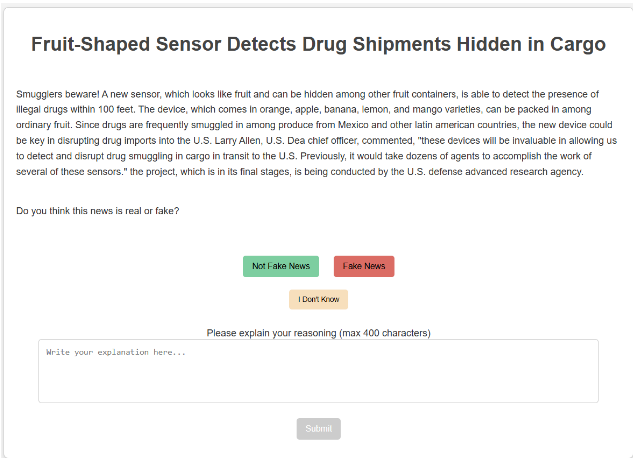

FIGURE 2. **Fake news detection task - *Evaluation Task*: task where participants had to classify each news article without the AI assistant. Done to understand carry-over effects.**

- **Linguistic complexity:** sentence structure, readability metrics.
- **Emotional valence:** sentiment polarity, affective language.
- **Cognitive markers:** certainty expressions, causal terms.
- **Social references:** pronoun usage, group identifiers.
- **Contextual anchors:** temporal references, named entities.

The features were extracted using established psycholinguistic lexicons (LIWC, MoralStrength) and NLP tools such as TextBlob and VADER [68], [69]. This process yielded over 100 interpretable indicators per article. Given our manually engineered tabular features, we selected XGBoost for three key reasons. First, extensive benchmarking shows tree-based methods consistently outperform deep learning on structured tabular data [70], while being more computationally efficient. Second, unlike neural networks, which require approximate post-hoc explanation methods, XGBoost can support exact post-hoc explanations, enabling precise and consistent feature attributions based on the model's structure [71]. Third, this aligns with misinformation research where interpretable models achieve competitive performance [28]. Additionally, the model's efficiency also enables real-world deployment on consumer hardware.

The model was trained and evaluated using the FakeNewsAMT dataset [72], a widely used benchmark for misinformation classification. The dataset contains 480 news articles, evenly split between fake and legitimate news, and labeled by human annotators. To prepare the data, standard pre-processing steps were applied: tokenization, stop-word removal, and contraction expansion, alongside spelling correction as described in the article. Feature selection was performed to reduce dimensionality and eliminate highly correlated attributes (threshold 0.9), yielding a smaller, more interpretable feature set. Recursive feature elimination and correlation analysis reduced the final input to about 20 core features. The selected features and their meaning are presented in Table 1.

TABLE 1. **Features and their meaning of the final AI detection system.**

Feature	Meaning
`readability_Kincaid`	Grade level required to understand the text
`readability_dir-speech`	Proportion of direct speech
`liwc_function`	Frequency of words that structure sentences (e.g., pronouns, prepositions)
`liwc_auxverb`	Frequency of auxiliary verbs (e.g., is, have, do)
`liwc_conj`	Frequency of conjunctions (e.g., and, but, or)
`liwc_verb`	Frequency of action or state words (e.g., run, be, feel)
`liwc_interrog`	Frequency of question-related words or structures
`liwc_quant`	Frequency of quantity-related words (e.g., many, few, some)
`liwc_social`	Frequency of words related to social interactions (e.g., friend, talk)
`liwc_family`	Frequency of words related to family (e.g., mom, father, grandson)
`liwc_insight`	Frequency of words indicating deep thinking or understanding (e.g., think, know)
`liwc_certain`	Frequency of words expressing certainty (e.g., always, never, definitely)
`liwc_differ`	Frequency of words indicating differences or contrasts (e.g., but, versus)
`liwc_see`	Frequency of words related to visual perception (e.g., see, look)
`liwc_hear`	Frequency of words related to auditory perception (e.g., hear, listen)
`liwc_sexual`	Frequency of words related to sexuality
`liwc_affiliation`	Frequency of words indicating connection or affiliation (e.g., group, team)
`liwc_relativ`	Frequency of words related to spatial, temporal, or relational positioning (e.g., near, soon)
`liwc_motion`	Frequency of words indicating movement (e.g., go, move, run)
`moral_fairness`	Frequency of words linked to fairness or justice-related concepts

The classifier used in our study achieves an accuracy of 75% on the test set, a performance well aligned with prior work [72], [73] [74]. Importantly, rather than aiming to develop a new state-of-the-art model, our research seeks to explore how explanation types and interactivity shape user decision-making. For this purpose, a moderately accurate system can be desirable: a perfectly accurate system would offer limited analytical value, as users would rarely encounter situations requiring trust calibration or independent judgment. A moderately accurate model, therefore, provides the appropriate conditions for studying reliance dynamics. As part of the model selection process, we evaluated five alternative classifiers on the same feature set and train/test split: CatBoost, LightGBM, Random Forest, Logistic Regression, and SVM. Tree-based ensemble methods (CatBoost, LightGBM, Random Forest) yielded comparable performance to XGBoost (71–72% accuracy), while linear models (Logistic Regression, SVM) performed notably worse (57–61%), consistent with the non-linear nature of the feature interactions in this dataset. XGBoost was selected as it achieved the best overall performance.

For the user study, we selected 13 instances from the model's test set: 8 for the Categorization Task and 5 for the Evaluation Task. These instances were randomly selected under constraints on readability (300-600 words), topical diversity, and AI classification accuracy. The final sample included a balanced mix of topics. In the Categorization Task, the AI system produced correct predictions in 6 of 8 instances, achieving a final accuracy of 75% to mimic our model's performance.

IV. EXPLANATION TECHNIQUES AND CONVERSATIONAL SYSTEM

We provide two types of explanations: feature-importance and counterfactual based, produced by established post-hoc methods and delivered either directly or through a constrained conversational assistant. We focus on these two explanation categories because they represent widely recognized forms of local interpretability and a complementary nature [75]: feature-importance explanations address the question of *which* aspects of the input most influenced a prediction, while counterfactual explanations address *how* the prediction could change under minimal modifications. Prior work [76] in human-centered XAI shows that users benefit from both descriptive accounts of model behaviour and actionable "what-if" reasoning, particularly in unfamiliar content such as news articles. Counterfactuals also generate valuable insights in tasks that require high-stakes business decision-making [77]. Moreover, these two categories offer well distinct semantics that can be translated into natural language, making them suitable for integration with a controlled conversational interface.

A. FEATURE-IMPORTANCE EXPLANATIONS

Feature-importance explanations are generated using TreeSHAP [71], which yields an additive vector of local attributions:

$$\phi(x) = (\phi_1(x), \ldots, \phi_m(x)),$$

quantifying each feature's marginal contribution to the prediction $b(x)$. TreeSHAP computes exact Shapley values for tree-based models by exploiting the structure of decision ensembles, guaranteeing local accuracy and consistent feature attributions. This ensures that the attributions sum to the model output for the given instance and that features contributing more to the prediction receive strictly larger Shapley values.

To increase interpretability and reduce cognitive load, we show only the Top-K feature attributions. We selected a threshold of K=5 to prioritize high-impact information while maintaining cognitive ease. This specific value was chosen because our analysis showed that the first five features on average account for more than 70% of the total attribution magnitude:

$$e_{\mathrm{FI}}(x) = \{(a_i, \phi_i(x)) : i \in \mathrm{TopK}(x)\}.$$

These are subsequently presented as a ranked, textual list, emphasizing the relative importance and direction (positive or negative influence) of each feature without exposing numerical values to avoid unnecessary cognitive load.

B. COUNTERFACTUAL EXPLANATIONS

Counterfactual explanations are obtained directly using LORE [78], a well-established method for counterfactual generation. LORE constructs a synthetic neighborhood $\mathcal{N}(x)$ around the instance x to be explained, via its genetic-algorithm sampling procedure and trains a local decision-tree surrogate c_x approximating b around x.

Formally, given a local decision tree surrogate c trained in instance x's neighborhood, with $y = b(x) = c(x)$, LORE extracts a factual rule:

$$r = p \rightarrow y$$

where p is the conjunction of split conditions along x's root-to-leaf path in c, and a set of counterfactual rules:

$$\Phi = \{p[\delta_i] \rightarrow y'_i \mid y'_i \neq y\}$$

where each $p[\delta_i]$ represents p updated with minimal changes δ_i satisfying:

$$\delta^* = \arg\min_{\delta} \; nf(p[\delta], x) \quad \text{and} \quad p[\delta^*] \models U$$

with $nf(p[\delta], x) = |\{sc \in p[\delta] \mid \neg sc(x)\}|$ counting falsified conditions, and U being actionability constraints.

The complete explanation is:

$$e(x) = \langle r, \Phi \rangle$$

From Φ, we extract our explanation and translate the counter-rule into short natural-language statements describing only the direction of required feature changes, omitting the actual values to avoid cognitive load on the user.

C. CONVERSATIONAL ASSISTANT

The conversational assistant serves as a controlled natural-language interface for delivering explanations to users. Its primary function is to faithfully paraphrase the pre-computed explanations from TreeSHAP or LORE without introducing new reasoning, extrapolations, or factual content. To ensure this, we implement a defensive prompting strategy with multiple layers of constraints.

The assistant is instantiated using GPT-4 [79] with its temperature parameter set to 0 to maximize determinism and reproducibility. For each interaction concerning a specific news article, the system constructs a context-rich system prompt containing the article text x, the model's binary prediction y, and the corresponding explanation $e_{\mathrm{FI}}(x)$ or $e_{\mathrm{CF}}(x)$. The assistant's core instruction is to generate a natural-language paraphrase conditioned *only* on this provided context:

$$\mathrm{Assistant}(x, y, e) \;\Rightarrow\; \mathrm{NL}(e).$$

To guarantee the faithfulness and safety of outputs, the prompt enforces the following constraints:

- **Explanation grounding.** The assistant is prohibited from generating any reasoning, feature attributions, or counterfactual statements not explicitly present in the tuple (x, y, e). The instructions specify: "Assist the user using only the context provided."
- **On-topic enforcement.** The assistant's domain is limited to discussing the verification of the specific news article. For off-topic queries, the assistant must respond with a canonical refusal message.

For each article, a dedicated chat context is maintained, scoped by a unique `news_id`. When the user moves to a new article, the full conversational context is reset. See the supplementary material for the details of both prompts used.

D. REWARD SYSTEM

Both the **Categorization Task** and the **Evaluation Task** incorporated a gamified reward system designed to simulate real-world stakes and introduce a sense of risk. This design choice was motivated by prior research emphasizing the role of risk and consequences in shaping human-AI trust and reliance behaviors [12], [56].

Participants earned a performance-based monetary bonus. For each correctly classified article, they received £0.08; for each incorrect classification, £0.04 was subtracted from their bonus. If a participant selected *"I Don't Know"*, neither a reward nor a penalty was applied. The total bonus could not drop below £0.

This structure was intended to encourage thoughtful decision-making and simulate cost-sensitive choices under uncertainty. It also allowed us to interpret reliance-related behavior more meaningfully. For instance, choosing *"I Don't Know"* despite a correct AI prediction may indicate a lack of trust, while agreeing with an incorrect AI prediction may suggest overreliance. These behaviors are further discussed in Section III-C. Participants were informed of the bonus structure in advance, and all payments were made in addition to a fixed base compensation of £4.

E. EXPERIMENTAL PROCEDURE

The experiment began with a brief introduction to the study, during which participants were informed of their rights and asked to provide informed consent for the collection of performance data. Participants were also given a concise overview of the task, including detailed explanations of the rules and the reward system. Following the initial briefing, participants proceeded to the Categorization Task, where they reviewed and categorized eight news articles with the assistance of the NewsBot AI system. Each article was presented individually, and the AI provided a classification suggestion along with an explanation, based on the assigned experimental condition. After completing the Categorization Task, participants moved on to the Evaluation Task, where they independently classified five additional news articles without any AI support. In this task, participants were asked to provide a brief explanation for each classification decision. For both the categorization and evaluation tasks, participants were not permitted to browse the web or use their smartphones in order to avoid biasing the results. To enforce this, participants were automatically timed out if they remained inactive on our webpage for more than two minutes while viewing a news article.

Upon completing the tasks, participants completed a post-task questionnaire that collected information about their AI usage habits and news consumption patterns. This data was used to build an AI and news consumption profile. Participants were also asked about their confidence in both the AI system and their own classification decisions, as well as their overall trust in the AI. Finally, participants received a performance summary detailing how many articles they had correctly classified in each task. Figure 3 provides a graphical representation of the experimental flow.

F. PARTICIPANTS

The study was conducted on Prolific, an online platform commonly used to recruit diverse participants for behavioral research. The eligibility criteria required participants to be at least 18 years old and fluent in English, ensuring they could fully understand the experimental instructions and the content of the news articles. To simulate a realistic scenario in which non-experts encounter news without prior knowledge of its veracity, we aimed to recruit individuals unfamiliar with the specific articles. Since the dataset primarily contained US-based content, we deliberately excluded US residents as a methodological control: familiarity with the specific political and media context could have conferred an uncontrolled advantage in the detection task and thereby threatening the internal validity of the study. Each participant received monetary compensation consisting of a fixed base amount plus a performance-based bonus (approx. £10.50/hr).

We determined the required sample size using an a priori power analysis with G*Power software [80], [81]. To detect a medium effect size ($f = 0.25$) in an ANCOVA design (numerator $df = 1$, four groups) with a statistical power of .80 and a significance level of $\alpha = .05$, a minimum of 128 participants was required. To account for potential exclusions, 192 participants were initially recruited. Exclusion criteria included: (1) navigating away from the experimental tab for more than two minutes, a measure applied upon the presentation of each news article to prevent external information seeking, and (2) exceeding the standard timeout thresholds programmed within the Prolific platform. After excluding sessions, 161 valid sessions remained, exceeding the required sample size for adequate statistical power. Among these participants, 93 identified as female and 64 as male, with an average age of 33 years ($SD = 11.22$). Participants were evenly distributed across the four experimental conditions: static system with feature importance ($n = 41$), conversational system with feature

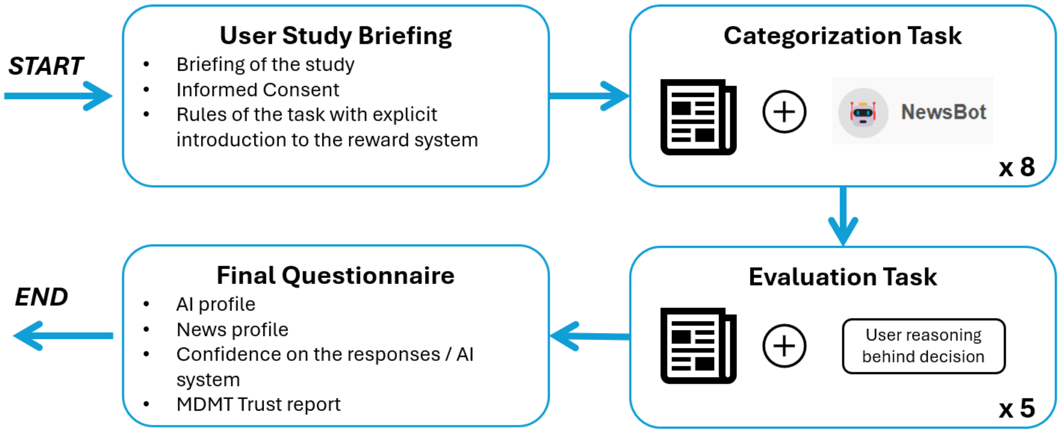

FIGURE 3. Flow of the user experiment.

importance ($n = 41$), static system with counterfactuals ($n = 39$), and conversational system with counterfactuals ($n = 40$).

G. MEASURES

The measures in this study fall into two main categories: dependent variables and covariates. The dependent variables capture participants' objective behaviour (e.g., task performance, AI reliance) and subjective perceptions (e.g. trust in AI). At the same time, the covariates are individual differences measured through a questionnaire and can also affect the fake news detection performance. Below is a detailed description of each set of measures.

1) DEPENDENT VARIABLES

a: OBJECTIVE TASK PERFORMANCE VARIABLES

To capture users' behavior and decision-making performance, we collected several objective task metrics. Some variables were measured in both the Categorization and Evaluation Tasks. In these cases, we analyze them either to assess direct effects of AI support (in the Categorization Task) or to examine potential carry-over effects once the AI is removed (in the Evaluation Task).

Accuracy Classification accuracy was measured for both tasks. For each participant, we separately computed the proportion of correctly classified news articles in the Categorization Task (with AI assistance) and the Evaluation Task (without AI). Accuracy in the former reflects immediate task performance with AI support; accuracy in the latter assesses carry-over effects on participants' independent decision-making.

Decision Time We recorded the time spent on each classification and calculated the average decision time per participant, separately for the Categorization and Evaluation Tasks. This measure helps assess cognitive effort across different conditions and whether exposure to AI assistance under those conditions affects efficiency in subsequent unaided decisions.

Agreement Rate In the Categorization Task, we computed the proportion of classifications that matched the AI system's recommendation. While agreement does not capture pre-AI intent, it is commonly used as a proxy for AI reliance. Higher agreement rates suggest greater alignment with the system's judgment.

Disagreement Rate Also measured in the Categorization Task, this reflects the proportion of decisions that explicitly contradicted the AI's suggestion (excluding "I don't know" responses). Given the presence of a no-penalty option, disagreement may signal confidence in one's own judgment and a deliberate override of the AI.

I don't Know Rate This measure reflects the proportion of times participants selected the "I don't know" option, available in both tasks. In the Categorization Task, it may indicate uncertainty or a lack of trust in the AI's recommendation. In contrast, in the Evaluation Task, it serves as a proxy for uncertainty when supporting evidence is absent. In both contexts, this option offered a low-risk alternative, making its use especially relevant for interpreting decision confidence and strategy.

We analyze Agreement, Disagreement, and "I Don't Know" rates as behavioral indicators of reliance. While subjective trust—measured through self-reports—captures the user's psychological attitude, these behavioral metrics provide a proxy for reliance within a realistic, single-exposure setting [65]. Specifically, the "I Don't Know" rate captures a state of "hesitant non-reliance", where the user recognizes the AI's input but lacks the confidence to either endorse or override it. By analyzing these three outcomes in tandem with our incentivized reward structure, we can distinguish between different reliance patterns which, by comparison, provide insights into the nuances of AI reliance for this specific task.

b: SELF-REPORTED VARIABLES

Human Confidence We assessed confidence through self-reported ratings on three items: participants were asked how confident they were in their classifications during the Categorization Task, how confident they were in their classifications during the Evaluation Task, and how confident they were in the NewsBot's recommendations. Each item was rated on a 5-point Likert scale. While the questions targeted slightly different aspects of confidence, they were internally consistent (Cronbach's $\alpha = .73$), so we averaged them into a single human confidence score. Higher values on this measure indicate greater overall confidence in both the participants' own judgments and the AI agent's suggestions.

MDMT Trust To assess trust in the AI assistant, we used the Multidimensional Measure of Trust (MDMT) questionnaire [82], which captures five key dimensions of trust: reliability, competence, ethics, transparency, and benevolence. Each subscale demonstrated strong internal consistency, with Cronbach's α values of .71, .90, .93, .91, *and* .94, respectively.

2) COVARIATES

In addition to the main dependent variables, two covariates were collected in the post-task questionnaire to account for individual differences that might affect participants' performance on the tasks:

AI Profile: To capture participants' prior experience with AI, we asked them to rate four items covering their frequency of AI use, familiarity with AI and machine learning tools, comfort with AI-based technologies, and general trust in AI systems. All responses were given on a 5-point Likert scale. The items showed good internal consistency (Cronbach's $\alpha = .79$). Therefore, we averaged them into a single AI profile score. Higher scores reflect more frequent use, greater familiarity, and generally more positive attitudes toward AI.

News Consumption Profile: This covariate assessed participants' news consumption habits, including how frequently they read news, how often they shared news on social media, their trust level in mainstream media, and their trust in news from social contacts. All responses were given on a 5-point Likert scale. The items showed an acceptable internal consistency (Cronbach's $\alpha = .69$). Therefore, we averaged them into a single News Consumption profile score. Higher scores reflect greater trust in News sources and more frequent news reading.

V. RESULTS

To analyze the effects of our experimental manipulations, we conducted a series of ANCOVAs with explanation type and system interactivity as between-subject factors. Participants' AI Familiarity (AI Profile) and News Consumption Profile were included as covariates to control for individual differences. Although some dependent variables used Likert or rating scales, they were treated as continuous, consistent with best practices when scale distributions approximate interval-level properties [83]. ANCOVA was preferred over non-parametric tests to enable covariate adjustment and more precise estimation of group effects.

To account for the increased risk of Type I errors due to multiple comparisons, we applied the Holm-Bonferroni sequential step-down procedure [84]. Significance thresholds were adjusted within six distinct conceptual families of tests: (1) the effects of the interactivity level factor on the categorization Task behavioral variables (RQ1); (2) the effects of the explainability type factor on the categorization task behavioral variables (RQ2); (3) the effects both explanation type and interactiivty level for carry-over variables (RQ3); (4) effects on self-reported measures; and (5) and (6) the influence of covariates on behavioral and self-reported measures, respectively. This approach ensures a rigorous control of the family-wise error rate (FWER) while maintaining sufficient statistical power to detect meaningful effects within each research hypothesis. Throughout the Results section, p-values that remained significant after this sequential adjustment are highlighted in bold.

In addition to these quantitative analyses, we conducted qualitative assessments to better understand how users engaged with the AI system. This included chatlog analyses in the conversational conditions and examination of participants' written justifications in the Evaluation Task. These insights helped contextualize the quantitative findings by revealing user strategies and reasoning patterns.

A. EFFECTS OF EXPERIMENTAL MANIPULATIONS ON CATEGORIZATION TASK

Results of the ANCOVAs for the categorization task are summarized in Table 2. Significant main effects of interactivity were found for the *Agreement Rate* and *Don't Know Rate* variables. Participants who interacted with the conversational system agreed more with the AI's recommendations and selected the "I Don't Know" option less often compared to those using the static system, regardless of explanation type. Participants in the conversational system condition also spent more time evaluating and deciding on each article compared to those in the static condition, though this effect did not survive Holm-Bonferroni correction. This result is expected, as the interactive nature of the conversational interface naturally encourages more extended engagement with the AI system. Main effects of explanation type were also observed. Specifically, participants who received counterfactual explanations showed lower *Accuracy*, longer *Decision Times*, and higher *Disagreement Rates* than those who received feature-importance explanations. Following Holm-Bonferroni adjustment for multiple comparisons, only the difference in *Decision Times* reached the required significance threshold. No significant interaction effects between explanation type and interactivity were found. Significantly, none of the covariates (AI Profile and News Consumption Profile) were significantly associated with any of the objective measures in the categorization task.

TABLE 2. Descriptive statistics (means and standard errors) and ANCOVA results comparing explanation types and interactivity levels.

Dependent Variable	Explanation Type					Interactivity Level					Covariates $F(p)$	
	Feature Imporance $M(SE)$	Counterfactual $M(SE)$	F	p	η_p^2	Static $M(SE)$	Conversational $M(SE)$	F	p	η_p^2	AI Profile $F(p)$	News Profile $F(p)$
Categorization Task (AI Assistance Effect)												
Accuracy	.64(.02)	.57(.02)	4.57	.030	.03	.59(.02)	.63(.02)	3.34	.070	.02	2.00(0.16)	0.12(.73)
Don't Know Rate	.064(.012)	.057(.011)	.40	.530	.01	.09(.013)	.03(.010)	14.02	**<.001**	.08	.74(.39)	1.32(.25)
Agreement Rate	.74(.02)	.67(.02)	3.74	.055	.024	.66(.02)	.75(.02)	10.81	**<.001**	.065	1.4(.24)	.39(.54)
Disagreement Rate	.20(.02)	.27(.02)	5.22	.024	.033	.25(.02)	.22(.02)	1.81	.18	.012	.55(.46)	.00(.98)
Decision Time	.75(.05)	.93(.06)	6.24	**.014**	.039	.75(.05)	.92(.05)	5.34	.020	.033	.12(.73)	.70(.40)
Evaluation Task (Carry-Over Effect)												
Accuracy	.63(.02)	.63(.03)	0.03	.860	.00	.59(.02)	.67(.02)	5.00	.027	.031	.24(.62)	.003(.96)
Don't Know Rate	.07(.012)	.09(.015)	1.735	.19	.011	.11(.02)	.05(.01)	10.885	**.001**	.066	.142(.71)	.32(.57)
Decision Time	1.74(.11)	1.91(.10)	2.109	.15	.013	1.79(.10)	1.85(.11)	.904	.34	.006	.844(.36)	5.615 (.019)
Self-Reported Measures												
MDMT Reliable	5.14(.14)	4.84(.16)	1.14	.29	.007	4.76(.17)	5.22(.13)	12.33	**<.001**	.074	9.48(**0.002**)	7.49(**.007**)
MDMT Competent	5.31(.17)	5.20(.18)	.001	.97	.00	5.27(.19)	5.25(.16)	1.14	.29	.007	10.88(**.001**)	4.913(**.028**)
MDMT Ethical	4.92(.21)	4.78(.20)	.052	.82	.00	4.80(.22)	4.91(.19)	2.21	.14	.015	6.63(**.011**)	7.08(**.009**)
MDMT Transparent	5.04(.20)	4.82(.19)	.16	.69	.001	4.94(.21)	4.93(.17)	1.3	.026	.009	10.75(**.001**)	6.54(**.012**)
MDMT Benevolent	4.74(.23)	4.52(.22)	.11	.75	.001	4.66(.23)	4.60(.23)	1.20	.27	.008	9.53(**.002**)	14.40(**<.001**)
User Confidence	3.62(.07)	3.37(.10)	2.08	.15	.013	3.56(.09)	3.44(.08)	.32	.57	.002	29.95(**<.001**)	12.41(**<.001**)

Note: Degrees of freedom = (1, 142). P-values that are bold are significant at threshold $\alpha = 0.05$ with Holm-Bonferroni correction.

B. CARRY-OVER EFFECTS ON EVALUATION TASK PERFORMANCE

To examine carry-over effects, we analyzed how prior exposure to the different experimental conditions in the categorization task influenced participants' performance in the subsequent evaluation task, which was completed without AI assistance. The ANCOVA results are also shown in Table 2. A significant main effect of interactivity level was found for *Don't Know Rate*. Participants who had previously interacted with the conversational system were less likely to select "I Don't Know" during the evaluation task. No significant main effects of explanation type were observed in the evaluation task. None of objective measures in the evaluation task showed significant associations with the covariates.

C. SELF-REPORTED PERCEPTIONS

The ANCOVA analysis revealed that both covariates, AI Profile and News Consumption Profile, were significantly associated with all self-reported perception variables. All subscales of the MDMT trust measure, as well as users' confidence ratings, were positively correlated with both covariates. Specifically, higher scores on the AI Profile and News Consumption Profile were associated with higher reported levels of trust (across all MDMT subscales) and greater confidence in both users' own decisions and in the AI system. These relationships are visualized in Figure 4, which displays the linear associations between the covariates and self-reported variables.

Notably, neither explanation type nor interactivity level of the AI system showed significant effects on the self-reported perception measures. In contrast, the covariates, participants' predefined AI Profile and News Consumption Profile, were more strongly associated with users' trust and confidence.

D. USER ENGAGEMENT WITH THE CONVERSATIONAL SYSTEM

To better understand how participants engaged with the conversational system, we analyzed interaction patterns by examining the number of interactions per participant, the length of prompts submitted, and the types of questions posed to the system. Participants using Feature Importance explanations generated 15% more interactions on average (Feature Importance: $M = 25.25$, $SD = 6.44$; Counterfactuals: $M = 21.38$, $SD = 3.74$). However, this difference did not reach statistical significance ($p = 0.19$). Feature Importance elicited slightly longer prompts, averaging 26.02 words ($SD = 4.03$) compared to 24.24 words ($SD = 5.26$) for Counterfactuals. This difference was also not statistically significant ($p = 0.49$).

We observed significant decreases in user interactions across both methods as participants engaged with later articles (Feature Importance: $\rho = -0.80$, $p = 0.017$; Counterfactuals: $\rho = -0.83$, $p = 0.010$). Nonetheless, prompt lengths remained stable over time. Figure 5 shows the number of total interactions and average prompt length over the news articles.

A simple qualitative analysis of the user-conversational system's chat logs was conducted to better understand the differences between explanations. Figure 6 shows this categorization regarding the frequency of each category in the two different explanation type conditions. This analysis revealed three primary themes of participant interactions: clarification requests ("Why Questions" and "Clarification/Explanations"): general questions aimed at understanding the reasoning behind the classification (e.g., "Can you explain this more?" or "Why is this fake?"). Challenge/verification queries ("Credibility" and "Speculation and Doubt" categories): prompts where users questioned or pushed back against the AI's suggestion (e.g., "I don't see how this is misleading. Are you sure?"). Content-oriented inquiries: requests focused on specific model attributes (e.g., "Which words made it seem fake?" - "Linguistics/Structure" category), as well as fact-checking requests ("Confirmation/Denial" and "Specific Aspect" categories). While both explanation types elicited comparable levels of content-oriented inquiries overall, participants in the counterfactual explanation condition asked significantly more clarification questions, particularly "why" questions, such as "Why is this classified as fake?" The results suggest that users may not intuitively grasp the contrastive nature of counterfactual explanations, which are designed to answer implicit "what if not?" questions by showing how minimal changes could alter the decision.

E. PARTICIPANT'S JUSTIFICATIONS IN THE EVALUATION TASK

A qualitative topic analysis was conducted to categorize participants' open-ended responses regarding the justification of their decisions (fake news or not fake news) in the evaluation task. One researcher developed the initial set of categories based on a preliminary review of the data. Two researchers then independently reviewed and coded all responses using the initial coding scheme. Through an iterative process, overlapping or redundant categories were refined, and responses were re-coded accordingly. Intercoder agreement was achieved for 65% of the responses. For the remaining 35%, discrepancies were discussed until a consensus was reached, resulting in a final, agreed-upon categorization of the responses. Each response could be assigned multiple codes. Table 3 presents the final categories, their definitions, and example participant responses.

The most common justifications provided by participants for classifying a news article as fake or not were plausibility assessment (32.8%) and source verification or prior topic-specific knowledge (27.9%). Participants tended to assess the veracity of the news based on whether the content appeared plausible, whether any sources were mentioned, or whether they had prior knowledge of the topic. Other frequently cited justifications concerned the information quality itself (15.3%), such as inconsistencies, levels of

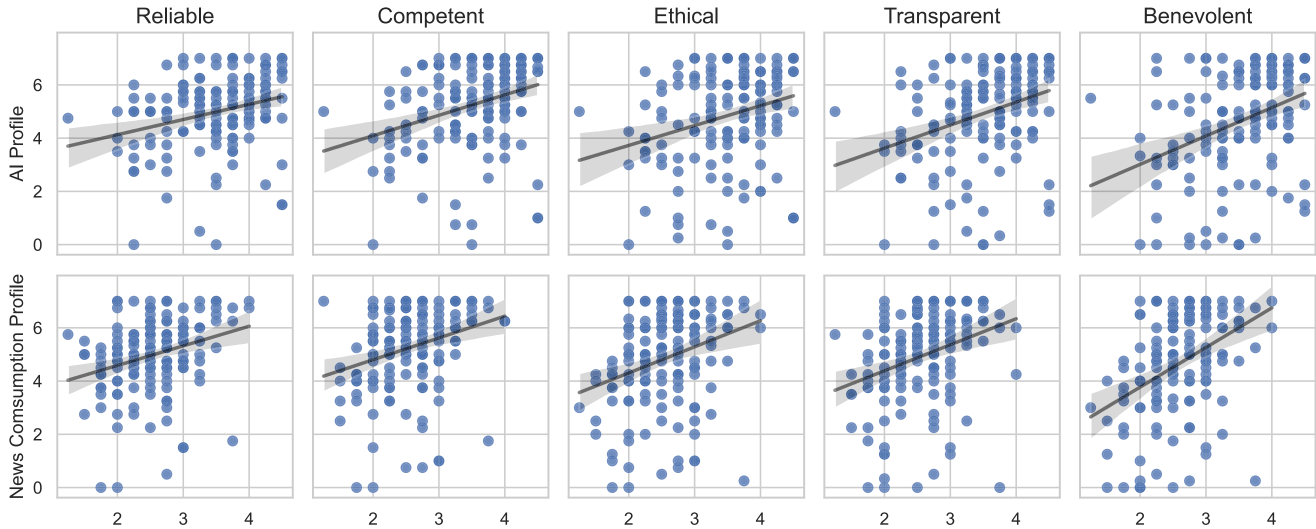

FIGURE 4. Scatterplots showing the relationships between self-reported dependent variables and the covariates (AI profile and news consumption profile). Each panel depicts a separate dependent variable, with trend lines representing linear fits. Shaded areas around the regression lines indicate 95% confidence intervals.

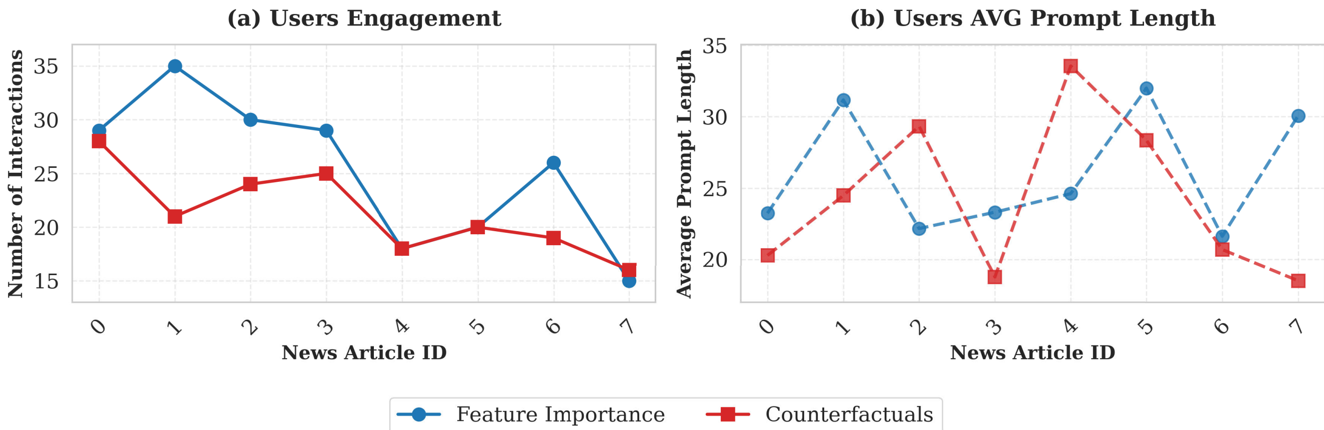

FIGURE 5. Relationship between user interactions and prompt length for feature importance and counterfactuals explanations. Both explanation types showed a clear decline in user interactions over time, while average prompt lengths remained consistent throughout the session.

detail, and missing information, as well as linguistic or stylistic cues, including spelling, grammar, and tone. Within the linguistic or stylistic category, we identified a specific subset of responses that resembled the AI explanation reasoning. These were grouped under a dedicated category labeled "SimilarAIExplanation". While 14.2% of all coded responses fell into the broader linguistic or stylistic category, only 5% were classified as similar to the AI's explanation. The distribution of all justification categories is shown in Figure 7.

To further examine whether participants adopted reasoning strategies aligned with those of the AI model, additional analysis focused on the *Similar to AI Explanation* category. Chi-square tests of independence were conducted to assess whether the presence of this justification type was associated with the experimental conditions, namely, the explanation type and the level of interactivity. A significant association was found between explanation type and the frequency of *Similar to AI* justifications ($\chi^2(1) = 10.90$, $p = .001$). Participants who received counterfactual explanations produced more *Similar to AI* justifications (30 responses) than those who received feature-importance explanations (9 responses). Additionally, a significant association was observed between interactivity level and this justification type ($\chi^2(1) = 4.50$, $p = .034$). Participants in the conversational system condition provided fewer *Similar to AI* justifications (13 responses) compared to those in the static system condition (26 responses).

VI. DISCUSSION

In this study, we examined how users interact with an AI system in a fake news detection task, a realistic and complex decision-making context. We investigated how the level of interactivity of the AI system influences user behavior and decision performance **[R1]**, and how different types of explanations affect the dynamics of human–AI collaboration **[R2]**. Additionally, we explored the carry-over effects of these design choices on subsequent decision-making tasks

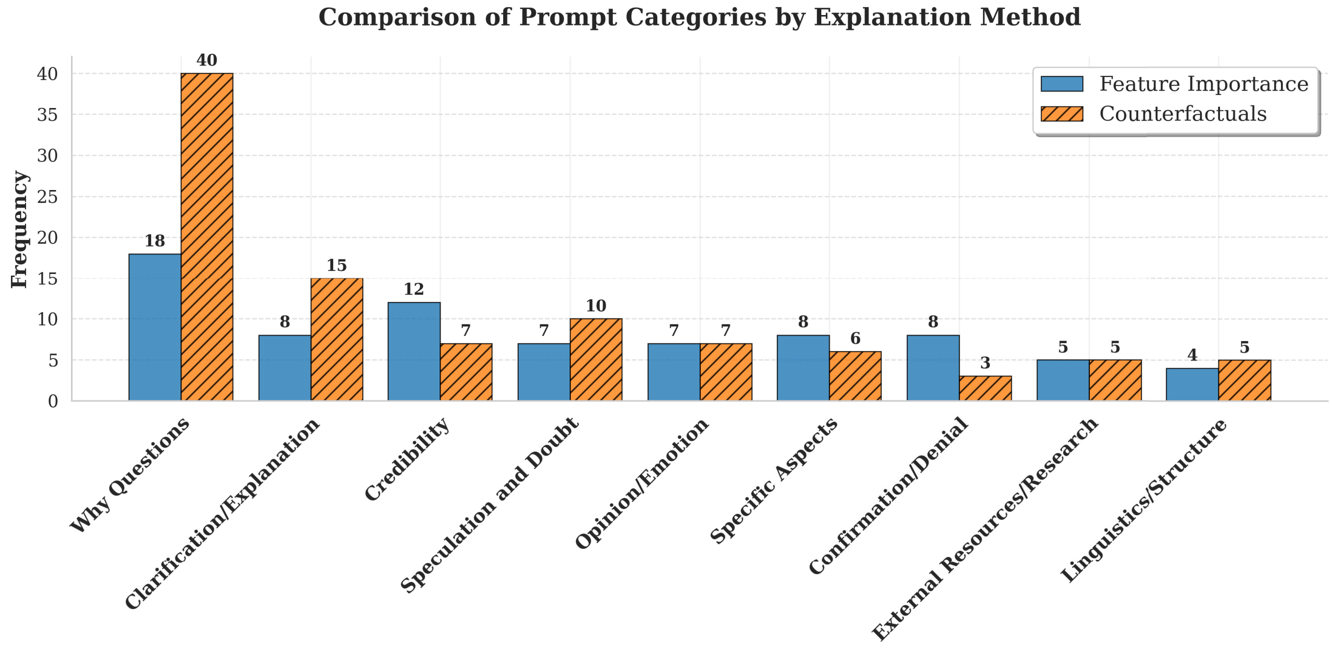

FIGURE 6. Comparison of user prompts in chatlog categories between feature importance and counterfactuals explanation types. the bar chart shows the frequency distribution of user prompts, highlighting differences in engagement patterns across the two explanation types.

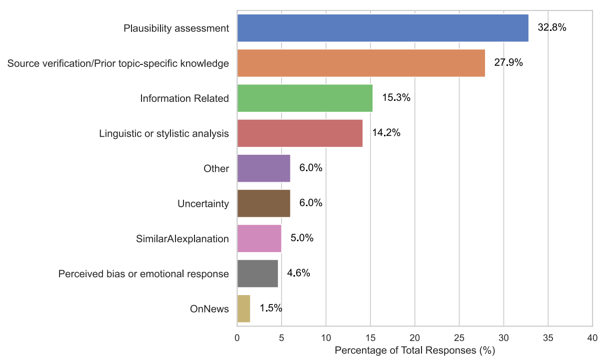

FIGURE 7. Percentage of categories in responses of decisions justifications (multiple categories per response allowed).

where users no longer have access to AI support **[RQ3]**. This is an important line of inquiry, as users in everyday life may encounter situations in which they must transition from assisted to independent decision-making. In the following sections, we discuss the findings of our study, organized around three key themes, and reflect on their broader implications for the Human–AI Interaction field and the fake news detection domain.

A. THE IMPACT OF CONVERSATIONAL SYSTEMS IN AI-ASSISTED DECISION MAKING

Our results show that participants using the conversational system exhibited significantly higher agreement with the AI's recommendations and reported lower uncertainty, measured by fewer ''I don't know'' responses, compared to those who received only static textual explanations [**RQ1**]. These two results combined reveal a greater reliance on the conversational AI system, with more agreements and fewer ''I don't know'' responses, suggesting a shift toward agreement with the AI system. However, this increased reliance did not translate into improved accuracy. In other words, while participants were more likely to follow the AI's suggestions when supported by a conversational interface, their overall decision performance remained unaffected, a common problem when reliance on the AI system is not calibrated, and users rely on the system when they should not [15], [85]. These findings raise important considerations for the role of conversational systems in AI-assisted decision making. Although conversational systems are often positioned as tools to promote deeper user engagement and reflection [14], [64], our study suggests that they may primarily serve a persuasive function, encouraging agreement with the AI, rather than enhancing understanding or improving outcomes. This aligns with emerging concerns that conversational systems might nudge users toward passive reliance rather than active sense-making [15]. This increased agreement with conversational AI systems is particularly problematic in contexts where users have low expertise and has been observed across several tasks involving LLMs [86], where people report reduced critical thinking when interacting with these agents.

Further, we observed a decrease in the number of interactions participants initiated with the conversational system over time [**RQ1**]. This declining engagement highlights a key challenge for the long-term effectiveness of conversational systems in AI-assisted decision making: the novelty of interactivity may wear off quickly, leading users to revert to more superficial or quicker interaction styles. Despite this trend, exposure to a conversational system appeared to have some carry-over effects. In the subsequent evaluation

TABLE 3. Final categories of participants' explanations for their decisions in the news evaluation task.

Category	Definition	Examples
Plausibility assessment	Reasons based on the inherent believability or perceived realism of the story's content. Users judge whether events seem likely, possible, or far-fetched given their world knowledge.	"It sounds like a plausible story"; "It just sounds a bit far fetched"
Source verification & Prior topic-specific knowledge	Justifications referencing external information, source checking, or relying on prior knowledge or beliefs about the event or topic. Includes references to specific domains (technology, law, celebrities, etc.).	"No credible sources"; "I believe it is not fake news because it is a common thing to read about autonomous vehicles nowadays."
Information Related	Focuses on the sufficiency, specificity, or lack of information in the article, such as the presence or absence of names, dates, locations, technical details, or conflicting information. Points out logical coherence, inconsistencies, or contradictions within the article itself.	"There is not enough information"; "There is a lot of detail to this story";"The structure does not sound right."
Linguistic & Stylistic analysis	Reasons based on language, tone, grammar, spelling, punctuation, capitalization, sentence structure, or writing style.	"The English is incorrect"; "Tone is not impartial"; "Poor grammar"
Perceived bias or emotional response	Reasons related to the user's emotional reaction or perception of bias, sensationalism, or intent in the article.	"It seems insulting"; "The use of language is high and provocative"
Uncertainty	Explicit statements where the user expresses an inability to judge due to a lack of knowledge or information.	"I do not know enough"; "I am not sure about this news"
Similar to AI	The user justifies their decision using linguistic or stylistic criteria similar to those highlighted by the AI system, such as direct speech ratio, use of conjunctions, or verb frequency.	"The frequency of conjunctions was lower"; "The proportion of direct speech was higher"
On News	Comments indicating whether the user has previously read or encountered the news story, or references to their familiarity with the news item.	"I've seen this news before"; "I haven't heard of this story before"
Other	Reasons that do not fit into any of the previous categories.	"No particular reason"; "Just a gut feeling"

task, conducted without AI support, participants who had previously used the conversational agent selected "I don't know" less frequently, suggesting increased confidence [**RQ3**]. The reason for this carry-over effect when users have access to a conversational system has to be studied further; however, we can hypothesize that having a conversational system to help them understand the AI reasoning gives the users more confidence that translates into their behaviour in subsequent decision tasks when they do not have an AI system to help.

Interestingly, while the system's design influenced behavioral measures such as agreement and uncertainty, subjective perceptions of trust and confidence were more strongly associated with individual differences, specifically participants' AI familiarity and news consumption profiles. It is important to note that self-reported trust and confidence were included as secondary outcomes in this study; the primary focus of our analysis remains on the behavioral metrics reported above. The limited effect of explanation type and interactivity on these subjective measures should therefore not be interpreted as a sign of measurement insensitivity. Rather, we argue that this constitutes a substantive finding: user perceptions of AI systems appear to be more strongly shaped by pre-existing individual dispositions than by interface design choices, a result that aligns with recent evidence in the human-AI collaboration literature [87]. This supports prior findings that behavioral and perceptual indicators of trust can diverge [88]. In our case, system-level features like explanation type and interactivity level shaped users' observable behaviors during the task, whereas users' pre-existing dispositions better predicted how they felt about the system. This finding demonstrates once again how HCI design choices can nudge user behavior and change human-AI collaboration dynamics, while users' subjective perceptions prove more resistant to change. Consequently, self-reported trust measures alone are insufficient proxies for behavioral reliance, and researchers should be cautious about conflating the two when evaluating AI-assisted decision-making systems.

Together, these results suggest that while conversational systems can alter how users interact with AI recommendations, they do not necessarily improve decision quality or human-AI collaboration. Looking ahead, it will be important to conduct additional research to understand how conversational systems, especially those powered by LLMs, can move beyond influencing reliance and begin to improve users' understanding and collaboration with AI.

B. THE ROLE OF DIFFERENT EXPLANATIONS IN THE DESIGN OF AI SYSTEMS

Our findings suggest that the type of explanation presented by an AI system can influence user behavior and task performance [**RQ2**]. Specifically, participants who received feature-importance explanations tended to be more accurate in the categorization task than those who received counterfactual explanations. While this difference in accuracy did not reach the threshold for significance after Holm-Bonferroni correction, the observed trend aligns with our qualitative data: participants often referred to or asked questions about specific features during their interactions with the conversational system, suggesting they found these explanations more interpretable and helpful.

By contrast, counterfactual explanations, despite their theoretical appeal and alignment with natural human reasoning in some domains [58], [89], appeared less effective in this context. Users took longer to make a decision when counterfactual explanations were provided and tended to disagree more with the AI recommendation than users with access to feature importance. These results imply that, given their contrastive nature, counterfactual explanations were more challenging to interpret in this context, where users are not experts. Presenting counterfactual explanations might be an effective way to show explanations if users are already aware of what features are more important to the classification and even what the effects of features are in the outcome, which is usually the case in contexts of high expertise, like in healthcare, but not the case for this specific context. These findings emphasize that explanation methods cannot be assumed to generalize across domains. Human-centered evaluation remains critical to determining what works in each setting.

Furthermore, although we expected repeated exposure to structured AI explanations to influence participants' reasoning, this was largely not the case [**RQ3**]. In the follow-up evaluation task, where participants completed classifications without AI support, only 5% of their justifications resembled the AI's explanatory style. Instead, most participants relied on plausibility heuristics, prior knowledge, and credibility judgments, rather than echoing the system's use of linguistic or structural features. This points to a persistent misalignment between the AI's explanatory model and users' mental models, a problem echoed in prior Human-AI Interaction work [90], [91]. These results point to a lack of research in the XAI field to first understand users' and stakeholders' needs, and only then tailor explanations to be more personalized or domain aligned [92].

Taken together, these findings highlight several challenges in explanation design. First, not all explanation types support performance equally, even if they are theoretically justified. Most of the explanation types that are increasingly being developed need to be empirically studied through user studies for each domain and decision task characteristic. Something that is currently not done. Second, users may engage with explanations behaviorally (e.g., through interactions or agreement), but that does not mean they internalize the system's logic if their mental model of how the classification is done is similar to the AI system's. Bridging the gap between AI rationales and human reasoning requires more than technical clarity; it also requires alignment with how people actually evaluate information in practice.

These results underscore the importance of evaluating explanation effectiveness not just by user performance or preference, but also by how well explanations support users' reasoning in independent tasks. Future research should continue to explore how explanation methods can be adapted or framed to better align with real-world cognitive strategies—particularly in domains such as misinformation detection, where uncertainty and subjectivity are high.

C. REAL WORLD IMPLICATIONS

This study offers important implications for the design of AI-assisted decision-making systems, particularly in domains where users have limited expertise. Across a range of decision tasks that users make daily, such as fake news detection, personal financial decisions, and personal healthcare recommendations, AI systems are increasingly supporting non-experts in complex and daily decisions. In such settings, explainability and interactivity alone are not sufficient to guarantee effective human–AI collaboration.

Our findings indicate that conversational systems, despite their potential to improve engagement, can lead to increased reliance without improving actual decision accuracy. While conversational interfaces may create the impression of helpfulness or collaboration, they can also induce over-reliance on AI systems. For low-expertise users, this persuasive effect may reduce critical engagement rather than enhance understanding. This is particularly concerning in the context of public discourse and misinformation, where the persuasive nature of a conversational agent can mask the underlying uncertainty of the information it provides. If a system presents incorrect claims with high conversational fluency, it may inadvertently bypass a user's skeptical filters, leading them to accept false information as a verified fact. Ethically, this suggests that designers must balance the desire for seamless interaction with the need to preserve user agency. In these settings, conversational systems have to be thought of as part of a more complex design where cognitive friction or interaction cues are used to ensure that reliance is calibrated to the system's actual reliability [9].

Moreover, this study highlights the need for explanations tailored to users' mental models, or, at minimum, capable of gradually reshaping them. Explanations may be technically valid but cognitively unhelpful if they do not align with how users naturally reason about the task. The findings reinforce that effective explainability is not only a technical challenge, but also a human-centered design challenge, requiring sensitivity to user cognition, task context, and interaction over time. These insights generalize to many real-world domains where end-users are expected to make judgments based on AI recommendations without extensive training or domain expertise. In such cases, explanation strategies must bridge the gap between model reasoning and user intuition, not widen it. This calls for new methods in personalized explanation design, user modeling, and longitudinal evaluation of user–AI collaboration.

In the specific context of fake news detection, these challenges become even more pronounced. The proliferation of AI-generated misinformation and increasing media polarization have made it essential to empower everyday users to assess the credibility of information. Nevertheless, our results show that even with access to AI recommendations and explanations, users rarely adopt the AI's reasoning strategies and often fall back on their own heuristics, such as plausibility, familiarity, or perceived source credibility. Despite the promise of XAI techniques to make AI decisions more transparent, standard explanation types like feature importance or counterfactuals do not automatically support better decision-making in this context. Feature-based explanations were more intuitive and led to higher accuracy, while counterfactuals often introduced confusion, suggesting that explanation types must be carefully matched to both the task and the target user.

In sum, these findings call for a more human-centered approach to designing AI support tools in disinformation detection: one that aligns explanatory models with user cognition, supports gradual learning, and mitigates over-reliance. More broadly, they underscore the need for HCI research to engage deeply with how explanation, interaction, and user expertise shape the dynamics of trust, reliance, and performance in high-stakes, low-certainty decision-making tasks.

D. LIMITATIONS AND FUTURE WORK

This study has several limitations that also highlight opportunities for future research.

First, the range of explanation types tested was limited. Future studies should explore a broader set of explanation strategies. The priority should be to include those which may be better aligned with users' mental models and intuitive reasoning processes, such as example-based, narrative, or uncertainty-based explanations. Understanding how different explanation formats impact comprehension and trust across diverse user groups is critical for advancing human-centered XAI.

Second, future work should employ larger article sets and longitudinal designs to investigate how decision-making dynamics evolve under extended exposure. Because our assessment of carryover effects was limited to an immediate follow-up task within a single session, it remains unclear whether these shifts in behavior and confidence persist over longer periods, such as days or weeks. Transitioning to longitudinal tracking would provide a more robust understanding of whether AI-assisted decision-making can foster lasting skills or if it results in more durable, long-term attitudes toward AI systems.

Third, this study was conducted in a controlled experimental setting. While this allowed for systematic evaluation of variables, it does not fully capture the complexity of real-world environments. Future research should investigate how fake news detection systems perform when deployed in more naturalistic contexts, such as social media platforms or news aggregation tools, where user attention, behavior, and external distractions may significantly affect interaction with the system.

Fourth, the AI model used in this study was based on feature-engineered text representations rather than a state-of-the-art deep learning architecture. While this choice was intentional to achieve more interpretable explanations, it may limit the generalizability of the findings. Future work could explore whether more advanced models, such as those using transformers or multimodal inputs, produce different results, both in terms of explanation quality and user outcomes. However, because explanations from deep learning models are often more abstract and more complex to understand, it remains a key challenge to make such systems interpretable and usable for end users.

Finally, the conversational interaction was relatively limited in scope. Future work could explore alternative or more proactive interaction strategies. For instance, conversational systems could be designed to actively prompt user reflection or encourage deliberation by asking follow-up questions, clarifying user doubts, or prompting users to justify their decisions. Such approaches may foster deeper cognitive engagement and reduce the risk of passive over-reliance on AI recommendations. Additionally, a limitation of our conversational condition is the potential confounding effect of the LLM's inherent language style and perceived fluency. While our defensive prompting strategy successfully restricted the assistant from generating ungrounded reasoning, it did not explicitly control for the natural, authoritative tone characteristic of models like GPT-4. As noted in prior work [93], highly fluent language can act as a persuasive mechanism, potentially increasing user confidence independently of the explanation's actual quality. Future research should disentangle these factors by explicitly manipulating the conversational agent's tone (e.g., confident versus hesitant) or comparing LLM-generated dialogue against rigid, template-based conversational responses. Furthermore, a deeper discourse analysis of the conversational chat logs, such as a turn-level categorization of question types or interaction strategies, represents a valuable direction for future research.

VII. CONCLUSION

This study explored how different explanation types and interaction designs in AI-assisted fake news detection systems shape user behavior, decision-making, and trust. By examining both in-task performance and downstream effects on user decisions without AI support, we provide new insights into the dynamics of human–AI collaboration in realistic low-expertise settings.

Our findings show that while conversational systems can reduce uncertainty and increase agreement with AI recommendations, they do not necessarily improve decision accuracy, raising important questions about reliance and user calibration. Similarly, feature importance explanations appeared more effective than counterfactuals in this context, highlighting the need for better alignment between AI explanations and users' mental models. Notably, user characteristics, such as AI familiarity and news consumption habits, were stronger predictors of trust and confidence than system design features, suggesting the value of personalized design.

These insights contribute to ongoing research on designing AI systems that not only provide explanations but truly support human decision-making. For domains like disinformation detection, where the cost of over-trust is high, careful attention must be paid to both how systems communicate and how users interpret and internalize those communications.

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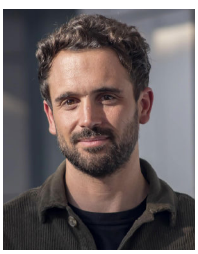

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